

Impact of mesoscale eddies on near-inertial wave vertical propagation in Iceland Basin

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August 19, 2025

Abstract

Near-inertial waves (NIWs) play an important role in ocean circulation and mixing through propagating and dissipating energy throughout the water column. Turbulent mixing of the ocean interior caused by NIWs notably modifies ocean stratification and contributes to the global redistribution of energy. However, the life cycle of NIWs from generation to dissipation depends on various processes interacting with the waves. The interaction of mesoscale currents with NIWs as well as the resulting modification of NIW kinetic energy and dissipation have been only sparsely observed. To quantify the impact of the mesoscale circulation on NIWs, we use an 18-month-long moored array from a region with enhanced winds and mesoscale activity (high eddy kinetic energy) in Iceland Basin. The moored observations reveal seasonal cycles in eddy kinetic energy and NIW amplitudes, as well as in the sign of depth-resolved relative vorticity. Local maxima of near-inertial kinetic energy are observed in a permanent pycnocline between 400 and 900-m depth from winter to early spring. The seasonal cycle of these deep energy maxima co-varies with seasonal cycles of wind stress and mesoscale anticyclonic circulation (negative vorticity). Downward propagation of NIWs in mesoscale anticyclones brings near-inertial kinetic energy into the permanent pycnocline. Trapped in this layer, NIWs partially dissipate with a time-averaged vertical profile of turbulent dissipation rate

peaking at $\sim 2.0 \times 10^{-9} \text{ W kg}^{-1}$ at 700-m depth. Dissipation time scales at depth match, on average, NIW event duration, pointing to locally generated turbulence in the permanent pycnocline as a key factor for terminating the NIW life cycle.

Significance Statement

This study explores how near-inertial waves (NIWs) and mesoscale currents work together to move energy from the surface to deeper ocean interior. Using over a year of observations in the North Atlantic, we found that kinetic energy from wind-driven internal waves are trapped by anticyclones leading to enhanced turbulent mixing in a permanent pycnocline. Our findings show that NIWs energy distribution and dissipation below the mixed layer is controlled by both seasonal winds and eddies. Understanding the vertical evolution of wind-driven kinetic energy and turbulent mixing in the ocean is key to improving numerical models, and thus for climate prediction.

1 Introduction

Internal waves are a crucial part of the global ocean as they play a role in energy pathways and mixing. Winds [Price, 1983, Alford et al., 2016, Alford, 2020], tides [Egbert and Ray, 2001, Melet et al., 2013], and interactions between background currents and bottom topography [Scott et al., 2011, Melet et al., 2014] can all generate internal waves. By nature, internal waves are sensitive to the environment in which they are generated and propagate. Temporal and spatial variations of the stratification such as pycnocline variability, fronts or eddies can impact their life cycle by leading to changes in their propagation path [Zhao et al., 2018b], modal distribution [Ponte and Klein, 2015, Buijsman et al., 2017] and dissipation [Whalen et al., 2020, de Lavergne et al., 2019]. Understanding these mechanisms and how internal waves are impacted by background variability is notably key to the improvement of numerical ocean climate models. In order to close energy budgets within these models, it is important to understand the theoretical underpinnings of the turbulent mixing parameterizations implemented to dissipate internal waves [MacKinnon et al., 2017, Whalen et al., 2020]. The geography of internal-wave driven mixing varies globally, dependent on regional patterns of internal wave breaking [Whalen et al., 2020]. Adding to the complex and variable geographic distribution, and

due to their small vertical scales relative to background currents, wind-driven near-inertial waves (NIWs) are prone to be modulated by background currents, notably anticyclones [Kunze, 1985]. Despite improving greatly over the last decade, our ability to simulate the spatial distribution of the energy and mixing associated with NIWs is still under-resolved and requires careful validation through the analysis of in situ and satellite observations [Arbic, 2022, Raja et al., 2022, 2024].

A notable interaction between the mesoscale and NIWs is the enhanced downward propagation of NIWs in anticyclones, i.e. eddies with negative vorticity [Kunze, 1985, Asselin and Young, 2020]. This is caused by the mesoscale vorticity shifting the NIW frequency to an effective inertial frequency, f_{eff} . Anticyclonic circulation favors downward propagation and trapping of near-inertial wave energy at depth where vorticity weakens. Initially referred to as the "inertial chimney effect", Lee and Niiler [1998] and Asselin and Young [2020] later suggested to coin it the "inertial drainpipe effect" to account for the enhanced downward propagation of waves. Both terms can still currently be found in literature. The theoretical description of the phenomenon is however limited as it is unable to predict the significance of the drainpipe effect on both the lateral and vertical distribution of NIW kinetic energy (NI KE), seasonal variability and associated dissipation rates.

Several studies have since endeavored to investigate the drainpipe effect further through observations [Thomas et al., 2020, 2024, Yu et al., 2022, Kunze et al., 2023, Shang et al., 2024] and modeling [Yang et al., 2023, Raja et al., 2022]. The spatial modulation of NI KE and dissipation by trapping of NIWs in anticyclones have been highlighted in realistic [Raja et al., 2022] and idealized [Yang et al., 2023] high resolution simulations. Focusing on one anticyclone and one wind event, both observations from floats [Thomas et al., 2020, 2024] and simulations [Yang et al., 2023] confirm the impact of the drainpipe effect in consistently enhancing the presence of NI KE at depth, below the mixed layer (ML). Event-based studies, limited to short periods of time (tens of days), do not enable conclusions on the impact of the drainpipe effect on longer time scales, e.g. seasonal [Thomas et al., 2020, 2024, Yang et al., 2023], and/or lack validation via in situ data [Yang et al., 2023, Raja et al., 2022]. Using longer observations (at least a year), the drainpipe effect has been observed in sub-mesoscale [Yu et al., 2022] and mesoscale anticyclones [Alford et al., 2012, Shang et al., 2024], showing that this effect plays a crucial role in bringing

NI KE below the mixed layer. The consequences of this means an enhanced propagation speed and dissipation rates at depth [Kunze et al., 2023]. The seasonality of the NI KE distribution was also shown to follow wind stress seasonality [Alford et al., 2012, Kunze et al., 2023].

The link between NIWs and mesoscale vorticity on seasonal time scales could not be achieved previously due to lack of horizontal resolution [Alford et al., 2012], focus on smaller scales [Yu et al., 2022] or limitations in extracting NIW signals [Kunze et al., 2023]. Based on observations from profiling EM-APEX floats in Iceland Basin, Kunze et al. [2023] could not find any consistent seasonality of the dissipation rates associated with NIWs. They argue this was likely due to their inability to correctly differentiate NIWs from semi-diurnal internal tides in their observations as they use moving observers (EM-APEX floats). Several questions thus remain open: How does the drainpipe effect modulate NIW energy when winds and stratification favorable to effective NIW generation coincide with enhanced presence of anticyclones? Do turbulent dissipation rates, and subsequent NIW-driven mixing, follow the same modulation? What is the impact on the stratification at depth? To answer these questions, we study the link between NI KE and dissipation rates and mesoscale vorticity over a period of more than one year at fixed points in an area of enhanced mesoscale anticyclone formation.

Iceland Basin is an area particularly favorable to the study of NIWs and mesoscale interactions, and the impact of these interactions on the seasonality of NIWs. Indeed, it is a well known site of internal waves: internal tides generated via interactions between surface tides and the Reykjanes Ridge [Vic et al., 2021] and NIWs due to frequent storms causing strong surface forcing [Klenz et al., 2022, Kunze et al., 2023]. Iceland Basin is characterized by the presence of deep Labrador Sea Water below Subarctic Intermediate Water and Subarctic Mode Water above 1000-m depth which sets up a complicated vertical structure to the stratification [Arhan, 1990, Bersch, 2002, Pollard et al., 1999, Zhao et al., 2018a]. Warm and salty water, forming a shallow seasonal pycnocline (10-50 m) in summer, give way to colder and fresher surface water in winter, accompanied by convection down to 500 m. A permanent pycnocline is also present around 500 to 800-m depth [Kunze et al., 2023]. A branch of the North Atlantic Current (NAC) flows from Southeast to Northwest in the upper layer in the Iceland Basin. This strong current flowing along the Rockall-Hatton Plateau slope triggers baroclinic instabilities leading to the formation

of long-lived anticyclones [Zhao et al., 2018a, Read and Pollard, 2001, Martin et al., 1998, Harris et al., 1997]. Trapping of NIWs in anticyclones has been observed in the area using short time series of shipboard observations over a few days [Thomas et al., 2020, 2024], and year-long time series from profiling floats [Kunze et al., 2023]. Despite the limitations associated with extracting NIW signals from the EM-APEX float observations deployed in Iceland Basin, Kunze et al. [2023] suggest a relationship between the trapping of NI KE in the permanent pycnocline and enhanced dissipation. However, the link between NIWs and potential vorticity remains to be identified as eddy formation in Iceland Basin follows variations on longer time scales due to slow changes of the NAC course [Read and Pollard, 2001, Martin et al., 1998]. Wind stress was the only factor identified as controlling NI KE seasonality by [Kunze et al., 2023]. The impact of mesoscale vorticity on NIW life cycle and seasonality, and implications for background features such as the permanent pycnocline, remain unknown.

This study aims to further investigate the life cycle of NIWs, especially the vertical distribution of NI KE, the vertical structure of dissipation rates, and their intra-annual variability. By using 1.5 years of moored observations, we are able to separate NIW signals from the semi-diurnal internal tides despite their proximity in frequency space. This allows the analysis of the seasonality of NIW kinetic energy and dissipation vertical structure with respect to both stratification and mesoscale vorticity. Seasonal high NI KE levels and dissipation rates are found in the mixed layer and at depth in the permanent pycnocline between 400 m and 900 m depth from late fall to early spring. We find that this seasonality is not only controlled by the wind stress seasonal cycle, but also by seasonality in mesoscale vorticity. High NI KE and dissipation rate are indeed observed in the permanent pycnocline only in anticyclones. We elaborate on the consequences of how seasonal variability of NIWs contributes to vertical mixing.

2 Observations and data products

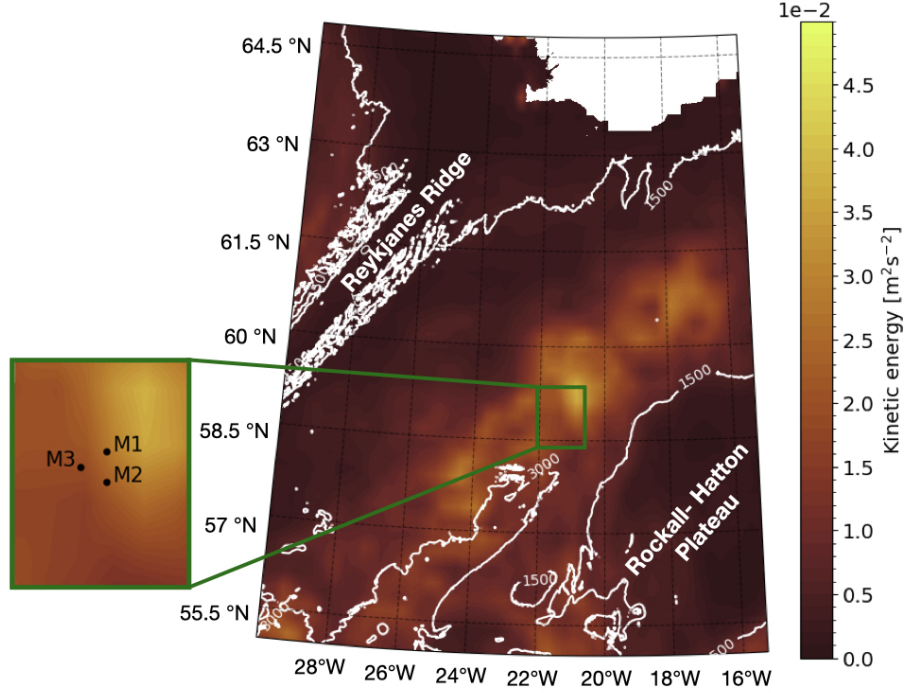


Figure 1: Iceland Basin topography and NISKINe mooring sites. Time-mean eddy kinetic energy derived from satellite altimetry from June 2019 to October 2020 (subset from CMEMS) is represented in the background. White contours correspond to bottom topography at 1500 and 3000-m depth. Mooring locations are indicated in the zoomed area (green rectangle) with black dots.

Iceland Basin is a region of enhanced eddy kinetic energy (EKE) throughout the year, visible in energy levels derived from sea surface height from satellite altimetry (Figure 1). It is also a region of internal wave generation through tidal (in the western basin; [Vic et al., 2021]) and seasonal wind forcing. The area, located between the Reykjanes Ridge and the Rockall-Hatton Plateau, is characterized by nearly flat topography with seabed depth around 2900 m.

2.1 Moorings

Three moorings were placed in a triangular array at the center of Iceland Basin in a region with enhanced time-mean EKE and away from any large topographic features as part of the Near-Inertial Shear and Kinetic Energy in the North Atlantic experiment

(NISKINe; Fig 1). A more comprehensive description of the moored instrumentation and the measurements made at M1 is provided in Voet et al. [2024]. The three moorings form a 15 km equilateral triangle centered around 59°6'N, 21°12'W. Moorings were deployed for 18 months between May 2019 and October 2020 measuring zonal and meridional velocity fields, u and v , over the upper half of the water column with ADCPs. The depth range of these measurements varies between the three moorings due to a few instrument failures and, at times, significant (>200 m) knockdown caused by deep-reaching energetic mesoscale currents.

Velocity measurements range between 75 m and 1500 m for M1, between 75 m and 1300 m with a gap between 150 m and 350 m for M2, and between 200 m and 900 m for M3 (Figure 3b-d). Horizontal velocity fields are gridded on a 16 m vertical grid for all moorings. Low-frequency horizontal velocities, $\bar{u}$ and $\bar{v}$, are obtained at each mooring by applying a low-pass filter with a Hamming window to detrended horizontal velocity fields with a cutoff frequency of 0.5 cpd and a window length of 15 days. EKE is estimated from the filtered velocity time series as $EKE = \frac{1}{2}(\bar{u}^2 + \bar{v}^2)$.

The spatial distribution of the NISKINe moorings was specifically designed to resolve mesoscale eddy shear at this latitude and thus allows estimates of horizontal gradient and vorticity, defined as

$$\zeta = \partial_x v - \partial_y u . \quad (1)$$

Müller et al. [1988] showed that estimates of relative vorticity could be obtained from only three data points by integrating (1) in a circle of radius R , passing through all points. The discretization of the integration gives

$$\zeta = \frac{2}{3R} \sum_{k=M1,M2,M3} [v_k \cos(\theta + \phi_k) - u_k \sin(\theta + \phi_k)] \quad (2)$$

where θ is the orientation of the array and $\phi_k = 4\pi/3, 2\pi/3, 0$. This formula is applied at every time step and over a depth range with velocity observations at all three moorings: 350 m to 900 m. The vorticity is then normalized by the local Coriolis frequency, $f \simeq 1.72$ cpd.

The inertial frequency at the mooring site is close to semi-diurnal, which complicates the extraction of the NIW signal. At the moorings, the semi-diurnal tidal signal is dominated by the M_2 harmonic of frequency $f_{M_2} \simeq 1.93$ cpd, only around 0.2 cpd away from

the inertial peak (see appendix, Figure A1). To limit leakage of the tidal signal into the inertial frequency band, we extract the near-inertial horizontal velocity fields ($\tilde{u}$, $\tilde{v}$) using a band-pass filter with a bandwidth of 0.2 cpd centered on f . The filter used is a finite impulse response filter designed using a Hamming window with 30-day length. The filter bandwidth was chosen to exclude the M_2 peak based on peak separation in time- and depth-averaged rotary spectra (Figure A1, blue curve). The risk of tidal signal contamination is lower during winter when the inertial peak dominates (Figure A1, orange curve). After filtering, gaps equal to or smaller than 12 hours in the $\tilde{u}(t)$ and $\tilde{v}(t)$ time series are interpolated with a 7th-order polynomial function. NI KE is then estimated from the filtered and interpolated horizontal velocity field as

$$\text{NI KE} = \frac{1}{2}(\tilde{u}^2 + \tilde{v}^2) . \quad (3)$$

Measurements of temperature, salinity and pressure via Sea-Bird SBE 37 as well as measurement of temperature via RBR SoloT and SBE 56 were sampled along mooring M1. SBE 37 were positioned at 47, 170, 316, 594 and 2850 m nominal depths. RBR SoloT and SBE 56 were more regularly placed, every 5 to 10 m. Buoyancy frequency N is calculated as

$$N = \sqrt{-\frac{g}{\rho_0} \frac{\partial \rho(z)}{\partial z}} \quad (4)$$

where ρ is density, which depends on temperature and salinity, ρ_0 is reference density, and g is the gravity constant. To understand the environment in which NIWs propagate, stratification is quantified during the deployment period, however, only temperature at M1 is available over the entire mooring depth range and deployment period. The seasonal cycle of climatological buoyancy frequency at the mooring site, denoted by N_{clim} , can first be obtained using both temperature and salinity data from the World Ocean Circulation Experiment (WOCE)-Argo climatology [Gouretski, 2019]. More accurate estimates of the local stratification, N , are obtained using temperature observations at M1 and salinity from the WOCE-Argo climatology (Figure 2e). Final corrections were applied to N when focusing on the effect of finer/shorter scale variations of stratification on NI KE distribution along the water column. Buoyancy frequency is thus derived from temperature and salinity provided by the SBE 37 when considering monthly variability of the stratification (see Appendix). Stratification estimates from mooring M1 corrected this way are referred to as $N_{corrected}$ in the rest of the study. Voet et al. [2024] estimated the mixed layer

depth (MLD) using a combination of temperature from mooring data and Argo MLD climatology (Figure 2e, blue curve).

2.2 Satellite and reanalysis products

A combination of satellite altimetry and reanalysis data products provide an essential overview of background mesoscale motion at the mooring sites. Low-frequency horizontal velocity fields at the surface are derived from Sea Surface Height (SSH) from altimetry [CMEMS] through geostrophic velocities, (u_g, v_g) ,

$$u_g = -\frac{g}{f}\partial_y SSH \quad (5)$$

$$v_g = \frac{g}{f}\partial_x SSH . \quad (6)$$

Surface EKE is then computed from (5) and (6) (Figure 1). Surface vorticity is derived from velocity inferred from altimetry by geostrophic balance following (1) with a horizontal resolution of $0.125^\circ \times 0.125^\circ$. The sparse satellite-derived vorticity is linearly interpolated to the center of the mooring triangle (Figure 2b, red curve) for comparison with products derived from moored observations (Figure 2b, blue curve).

Wind speed from 10-m above the sea surface at a 30-km horizontal resolution from ERA5 reanalysis [Hersbach et al.] is interpolated to the center of the moorings. Wind stress τ at M1 is estimated as

$$\tau = \rho_a C_D \vec{u}_r |\vec{u}_r| \quad (7)$$

where ρ_a is air density, C_D the drag coefficient, and $\vec{u}_r = \vec{u}_{10} - \vec{u}_s$ the relative wind speed, defined as the difference between 10-m wind speed, $\vec{u}_{10}$, and ocean mixed layer velocity, $\vec{u}_s$. The wind work generated in the mixed layer by the wind stress in the inertial frequency band assesses the amount of energy potentially going to near-inertial waves. Voet et al. [2024] provided such estimates of wind work at mooring M1, following

$$\tilde{\Pi} = \vec{\tau}_f \cdot \vec{u} \quad (8)$$

where τ_f is the inertial wind stress obtained by band-pass filtering around the inertial frequency.

3 Regional properties

3.1 EKE

Satellite-derived surface EKE and vorticity are comparable to their counterparts derived from the moored observations. The wide depth range available at mooring M1 captures the vertical structure of EKE in the upper 1400 m (Figure 2d). EKE is intensified in the upper layers, i.e. above ~ 700 m, with values approximately 2 times higher in this layer compared to what is found below. While active all year, EKE is higher during spring and summer (May to September) with a mean of $\sim 2.2 \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$ as opposed to $\sim 1.8 \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$ for the rest of the year. The depth of the intensified EKE maximum, i.e the depth at which EKE vertical gradient increases, does not have the same seasonality. It deepens slightly between September 2019 and March 2020 by approximately 50 m.

3.2 Vorticity

Satellite-derived vorticity, which lacks information in the vertical, is compared with the mooring-derived vorticity. Both surface vorticity from altimetry and depth-integrated vorticity from the moorings highlight similar seasonal cycles (Figure 2b). Vorticity is positive during summer (June to August) and mostly negative (anticyclonic) for the rest of the year with smaller absolute values on average ($\sim 8.1 \times 10^{-2}$ in summer; $\sim 6.4 \times 10^{-2}$ the rest of the year). The similarities between surface and depth-integrated estimates suggest a uniformity of the vorticity vertical structure, at least from the surface to 900-m depth. This is confirmed by the time-depth structure of the mooring-derived relative vorticity (Figure 2c). Depth independence is particularly noticeable during summer when vorticity is mostly positive (cyclonic). However, some decrease of absolute vorticity can be observed during the rest of year at 700-m depth ($\sim 7.6 \times 10^{-2}$ on average above 700-m; $\sim 5.7 \times 10^{-2}$ below). This coincides with the depth of maximum EKE vertical gradient (Figure 2d).

3.3 Stratification

In addition to strong seasonal mesoscale activity, stratification in Iceland Basin can also impact the propagation of NIWs by reflecting NIWs rays or impacting energy transfers

and mode distribution. The water column is highly stratified ($N \simeq 2.2 \times 10^{-5} \text{ s}^{-1}$ on average) close to the surface during summer and early fall. However, maxima are also observed between 400 and 900-m depth all year (Figure 2e). This layer corresponds to the separation between warm and salty water and the colder and fresher water below. This *permanent pycnocline* is a constant feature in Iceland Basin [Kunze et al., 2023] and is also observed in the WOCE-Argo climatology all year long (not shown). In the 18-month time-mean buoyancy frequency profile at mooring M1, the permanent pycnocline is characterized by an amplitude of $N \simeq 2.3 \times 10^{-3} \text{ s}^{-1}$ uniform over the depth range between ~ 450 m and ~ 850 m. It is noteworthy that this depth range coincides with the observed decrease in mesoscale EKE and vorticity. Above the permanent pycnocline, the depth of the surface mixed layer has a strong seasonal cycle. The mixed layer is shallow (around 55 m deep) from June to October but then deepens reaching 500-m depth in March 2020. While the permanent pycnocline remains all year long, its depth range and amplitude also vary seasonally. In August 2019, the buoyancy frequency of the permanent pycnocline was greater than $N \simeq 2.3 \times 10^{-3} \text{ s}^{-1}$ between ~ 440 and ~ 850 -m depth while in March 2020 the same levels were exceeded only between 620-m and 830-m depth (Figure B2).

3.4 Surface forcing

Surface wind stress is a major forcing mechanism for local generation of NIWs which are thus impacted by the seasonal variability of the winds. Bursts of high wind stress ($\tau \geq 1 \text{ N m}^{-2}$) appear between September 2019 and April 2020 (Figure 2a). Peaks in τ are also observed during summer, but they do not exceed 0.8 N m^{-2} . Klenz et al. [2022] found that ERA5 estimates used here might underestimate true wind stress by up to 40%. However, this uncertainty in wind magnitudes does not affect their seasonal cycle. Seasonality of the wind stress results in seasonal wind work in the mixed layer which also peaks between September 2019 and April 2020 [Voet et al., 2024].

3.5 Regional synthesis

The combination of seasonally varying low-frequency motion, stratification, and winds makes the NISKINE mooring area a particularly favorable location for the study of NIW generation and propagation into the ocean interior. High wind stress in fall and winter results in high wind work driving enhanced near-inertial velocity in the mixed layer.

292 During the same period, EKE, though lower than during summer, is high and vorticity is
 293 mostly negative. The superposition of these conditions may lead to amplified downward
 294 NIW propagation. The background mesoscale and stratification at mooring M1 also
 295 highlights a layer corresponding to a permanent pycnocline and the downward decrease
 296 of mesoscale EKE and absolute vorticity. The combined effects of these features and their
 seasonality on NIW life cycles are investigated next.

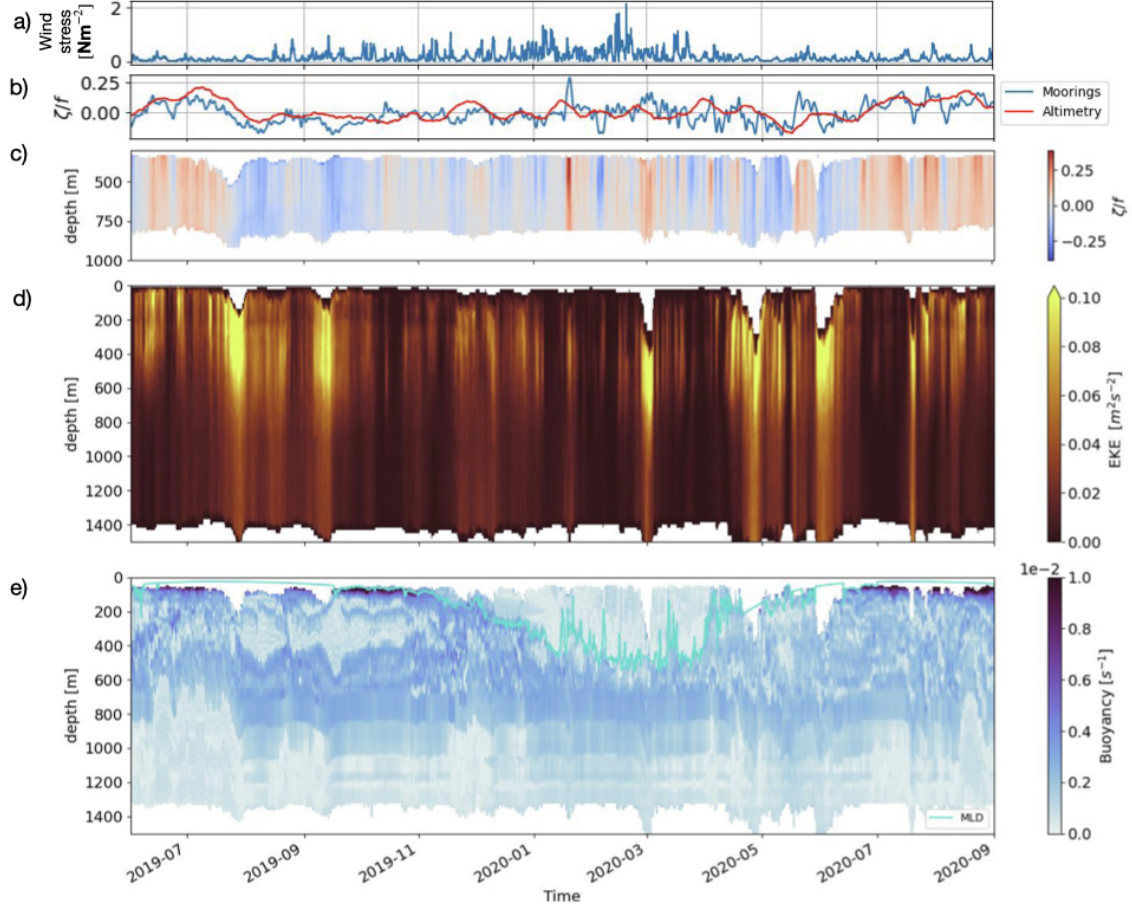


Figure 2: Iceland Basin properties at mooring M1. (a) Wind stress derived from ERA5 10-m wind speed. (b) Normalized relative vorticity at the surface from satellite altimetry (red curve) and depth-averaged estimate from the moorings (blue curve). (c) Normalized relative vorticity estimates from the moorings between 350-m and 900-m depth. (d) EKE estimated from low-pass filtered horizontal velocity time series at M1. (e) Buoyancy frequency N estimated from temperature at M1 and salinity from WOCE-Argo climatology. The blue curve represents MLD estimated from M1 observations and climatology.

4 Results

4.1 Near-inertial kinetic energy vertical distribution

Several events of elevated NI KE ($\geq 9 \times 10^{-3} \text{ m}^2 \text{ s}^{-2}$), separated by a few days to weeks, are visible in the mixed layer at M1 between September 2019 and June 2020 (Figure 3b-d). Similar patterns in NI KE are observed in the available depth ranges at M2, close to the surface, and at M3, during winter 2020 when the mixed layer is within the observation range of M3. The winter of 2020 is characterized by high wind stress, dominantly negative vorticity and deepening of the MLD (section 3). The high NI KE events correspond to local generation of inertial oscillations in the mixed layer by high wind stress at the ocean surface. Patterns of high NI KE within the mixed layer occur independently of the variations in mixed-layer depth, including the time period of the late winter mixed-layer deepening (February-March 2020; $> 500 \text{ m}$ MLD). This has not been reported previously as deep mixed layers typically result in small inertial responses, since typical slab models divide the energy by the MLD [Pollard and Millard, 1970, Kunze et al., 2023]. In our case, wind work is strong during winter 2020 at mooring M1 when the mixed layer reaches its maximum depth [Voet et al., 2024], enough to generate downward propagating NIWs. This was not observed by the profiling EM-APEX floats [Kunze et al., 2023], most likely because of the difficulty in separating near-inertial waves from semi-diurnal internal tides and the lower time resolution available from their profiling floats in late winter.

Enhanced NI KE is not only constrained to the mixed layer. High NI KE at depth (up to $\sim 1.6 \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$ at M1) is observed at all moorings with the same order of magnitude as in the mixed layer (maximum at $\sim 2.8 \times 10^{-2} \text{ m}^2 \text{ s}^{-2}$ at M1). Most events at depth appear to be locally wind-generated NIWs, initiating at the surface and then propagating below the mixed layer. The layer between 500-m and 1000-m depth stands out as particularly energetic, seemingly containing a large part of the NI KE signal below the mixed layer with values as high as those found close to the surface for some events (September 2019 and January 2020 are striking examples). This layer is characterized by a decrease with depth of the time-averaged EKE and absolute vorticity profiles as well as the presence of the permanent pycnocline (section 3, Figure 2c-e). The observed vertical distribution of stratification and mesoscale vorticity are likely determining factors for the NI KE distribution with depth. We aim to disentangle each feature's impact in the next

sections.

Focusing on a particular event illustrates possible mechanisms leading to the presence of enhanced NI KE at depth. In November 2019, NI KE is observed simultaneously at each mooring with no apparent local differences in the (coarsely) resolved surface vorticity and stratification (Figure 3, right panels). The MLD is shallow (150 m) and the maximum in NI KE ($\sim 10^{-2} \text{ m}^2 \text{ s}^{-2}$) is found in the mixed layer at M1 and M2. M3 observations do not cover the mixed layer during this period. Below the mixed layer, the depth of the NI KE maximum increases with time until it stops at 550 m at M1 and 650 m at M3. At M2, the NI KE signal below the mixed layer is not as visible. During this event (November 11 to November 24), negative vorticity is observed at all three moorings (Figure 3a, right panel). This example illustrates how anticyclones, through the drainpipe effect, enhance the downward propagation of NIWs and trap NI KE in the permanent pycnocline. Next we will show how seasonality in mesoscale vorticity acts to control the amount of locally generated NI KE getting trapped in the permanent pycnocline.

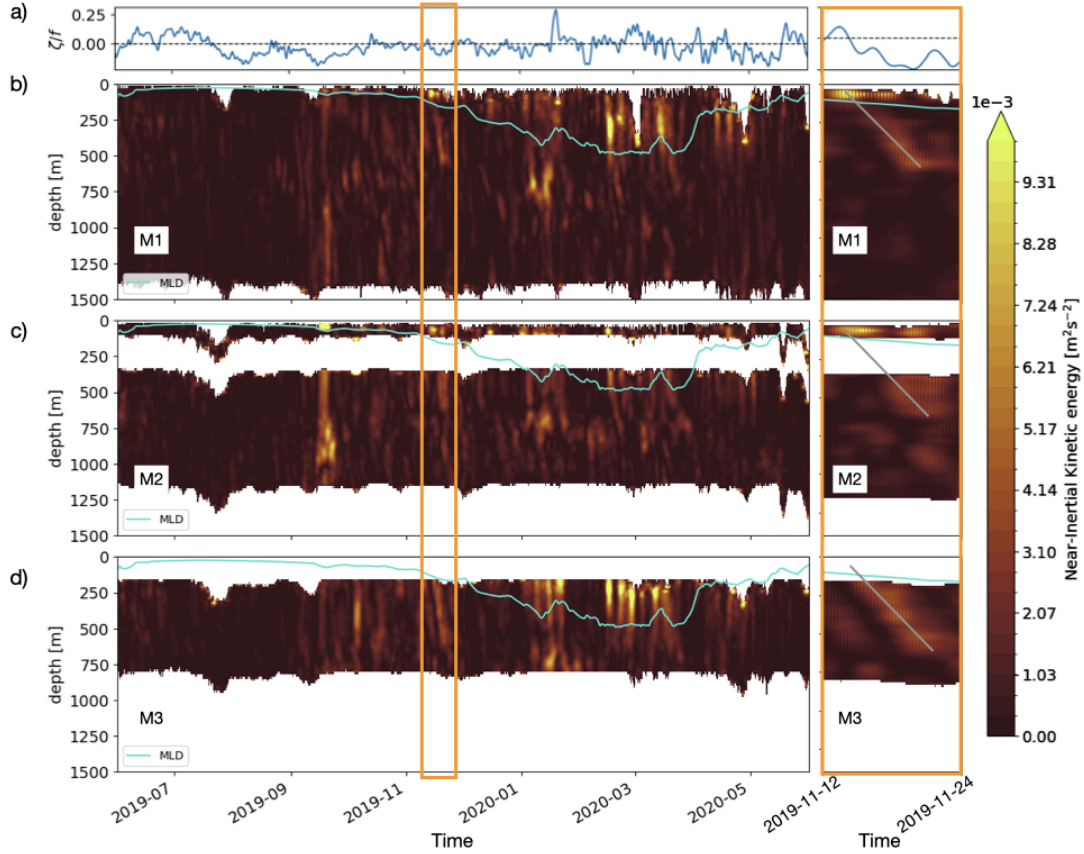


Figure 3: Normalized relative vorticity and NI KE at mooring sites. (a) Depth-averaged normalized relative vorticity from moored observations. (b-d) NI KE at mooring sites. MLD, smoothed using a moving average, is represented by the blue curve. (Left) Vorticity and near-inertial kinetic energy between June 2019 and June 2020. Orange rectangle represents the zoomed period in the right panels. (Right) Zoom on NIWs generated between November 12th and November 24th, 2019.

4.2 Trapping of downgoing near-inertial waves and links to vorticity

Focusing on downgoing NIWs reveals a connection between locally generated NIWs that enhances deep NI KE resulting from variability in vorticity and stratification. The direction of vertical energy propagation can be investigated by examining whether vertical shear follows a clockwise or counterclockwise motion with depth. These are associated with downgoing and upgoing vertical shear (i.e. downward and upward energy propagation), respectively. To estimate shear directionality, we calculate rotary spectra of velocity shear following previous studies [Gonella, 1972, Leaman and Sanford, 1975, Waterhouse et al., 2022]. This method consists in computing rotary spectra of horizontal velocity shear (u_z, v_z) and integrating over positive wavenumbers (clockwise) and negative wavenumbers (counterclockwise). Waterhouse et al. [2022] and Voet et al. [2024] provide more details about the method and its use to describe patterns of internal wave propagation. Following wind-driven NIW generation, downgoing shear variance dominates over upgoing shear variance (not shown, Voet et al. [2024]) and agrees with downward NI KE propagation from the mixed layer to the permanent pycnocline depth range (Fig. 3). Over the entire time series at all moorings, downgoing shear variance increases with NI KE (Figure 4, left panel). The same patterns in NI KE are observed whether only one mooring or all moorings are taken into account, indicating no local dependency (Figure 4, left panel, scatter plot for M1 only vs. black contours for all moorings). No correlation between significant upgoing shear and high NI KE events is observed. Previously, upgoing shear associated with high NI KE had been observed on jumps in stratification [Thomas et al., 2024]. Investigation of isolated NIW events with additional observations may bring such occurrences to light in a future study as discussed in section 5.

Enhanced NI KE below the mixed layer is associated with a ~ 500 -m MLD, i.e. from January to April 2020 (~ 2.5 more NI KE at depth in this period than for the rest of the year; Figure B3). The consistent presence of negative vorticity combined with enhanced wind power input in the winter facilitates the presence of high NI KE below the mixed layer. Downgoing shear variance and NI KE are higher when averaged over time and depth where vorticity is negative ($\sim 8.7 \times 10^{-8} \text{ m}^2 \text{ s}^{-4}$ and $\sim 6.3 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$) than when averaged for positive vorticity ($\sim 7.6 \times 10^{-8} \text{ m}^2 \text{ s}^{-4}$ and $\sim 5.4 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$). The vertical distribution of time-mean NI KE is also different depending on the vorticity sign (solid

375 and dashed black curves in Figure 4, right panel). A peak of NI KE appears at depth
 376 exclusively when vorticity is negative, with a maximum of $6.7 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$ around 675-m
 377 depth (Figure 4 right). In contrast, the time-mean NI KE profile is constant with depth
 378 (around $3 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$) for positive vorticity (dashed black curve in Figure 4 right). The
 379 vertical distribution of NI KE averaged over times of negative vorticity is also linked to a
 380 depth range characterized by high time-average stratification ($\sim 2.5 \times 10^{-3} \text{ s}^{-1}$). NI KE
 381 accumulates at the bottom of this layer, just above the decrease in stratification, around
 382 850-m depth. This depth range also coincides with the weakening in time-mean negative
 383 vorticity (blue curve in Figure 4, right panel). The increased number of anticyclones in
 384 winter combined with the presence of high stratification at depth allows for downward
 385 propagation and trapping of NIWs in a fixed layer below the ML.

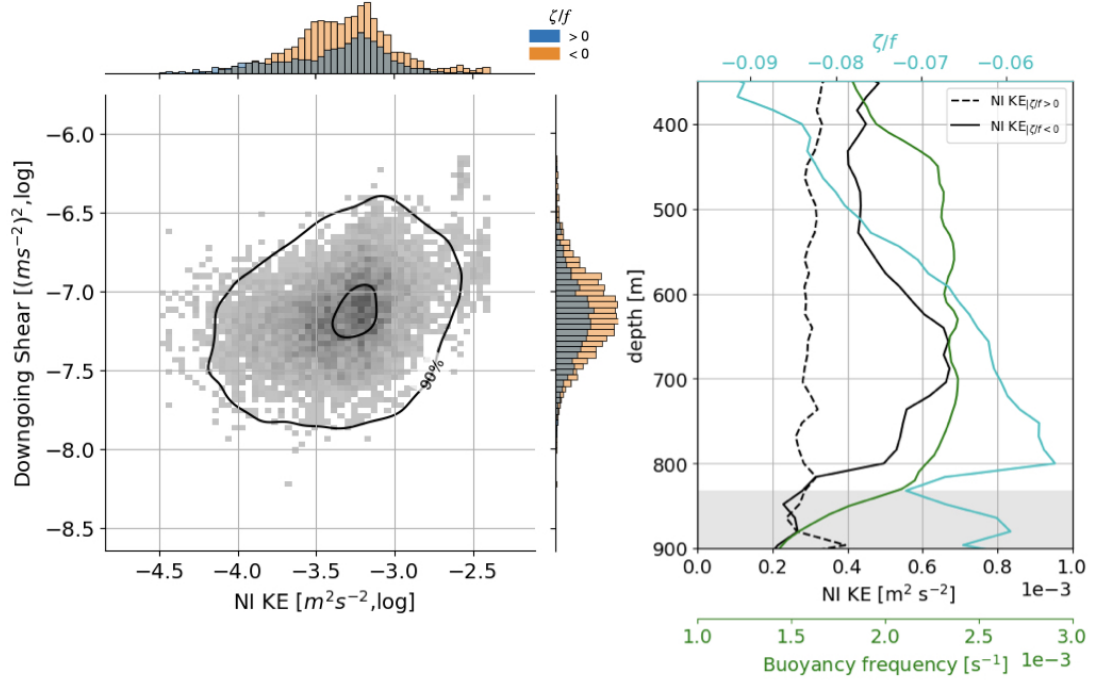


Figure 4: NIW vertical propagation and its link to mesoscale vorticity and background stratification at M1. (Left) Distribution of NI KE (x-axis) and downgoing shear variance (y-axis) at M1. Darker shades correspond to higher probability density. NI KE and downgoing shear variance distribution histograms are also shown depending on vorticity sign (orange represents negative vorticity and blue positive vorticity) in the margin plots. Black contours correspond to NI KE and downgoing shear variance probability density levels, estimated using all moorings. 90% and 10% levels are shown. (Right) Time-mean vertical profiles between 350 m and 900 m depth. Averaging period is the entire mooring time series. NI KE (black) is averaged for negative (solid) and positive (dashed) vorticity. Time-averaged buoyancy frequency N (green) and negative vorticity (blue) are also shown. The gray area corresponds to the depth range in which profiles were fewer, only half or less than the number available at the other depths.

4.3 Dissipation of NIWs

Individual NIW events generated locally in the mixed layer can be tracked through high NI KE down to the permanent pycnocline but no further, indicating a limit to downward propagation of NIWs. NI KE patterns below 850 m become less organized and uncorrelated with locally generated NIWs. There are three main explanations for this sudden decrease of NI KE with depth: 1) dissipation through turbulent mixing, 2) energy transfer (to low-frequency mesoscale or through wave-wave interactions), or 3) horizontal propagation / advection. The last option can be linked to the horizontal propagation of NIWs outside of the mooring site in the permanent pycnocline depth range by trapping and advection of NIWs by an anticyclonic eddy [Thomas et al., 2020]. Given the limitation of our available data sets, we focus in this study on the first mechanism: dissipation of NIWs through turbulent mixing. Weakening of NI KE at the mooring site by other means as well as ways to investigate them are discussed in section 5.

Turbulent dissipation, and the resulting diapycnal mixing, are typically inferred from observations over small vertical scales $\mathcal{O}(1)$ cm. Such measurements are not available from the moorings with velocity time series at 16 m vertical resolution. However, dissipation rate (and associated diffusivity) can be inferred using the finescale parameterization applied to observed vertical shear. Vertical shear spectra are calculated from horizontal velocity and integrated over vertical wavelengths ranging from 64 m to 600 m (Figure B1). The shear variance, $\langle V_z^2 \rangle$, is obtained from integrated spectra and compared to reference shear variance from the canonical Garrett and Munk (GM) model, $\langle V_z^2 \rangle_{GM}$ (Garrett and Munk [1975], Munk [1981], Gregg and Kunze [1991], Figure B1 right). Shear spectra are integrated over the same wavenumber range for both $\langle V_z^2 \rangle$ and $\langle V_z^2 \rangle_{GM}$. Dissipation rate of turbulent kinetic energy ε is estimated as

$$\varepsilon = \varepsilon_0 \frac{\overline{N_{clim}}^2}{N_0^2} \frac{\langle V_z^2 \rangle}{\langle V_z^2 \rangle_{GM}^2} h(R_\omega) j(f/N) \quad (9)$$

where $\varepsilon_0 = 7.8 \times 10^{-10} \text{ W kg}^{-1}$ is the reference dissipation rate [Waterman et al., 2014], N_0 the mean stratification and $\overline{N_{clim}}$ the time-mean stratification. R_ω is the shear-to-strain variance ratio,

$$R_\omega = \frac{\langle V_z^2 \rangle}{\overline{N} \langle \xi_z^2 \rangle}, \quad (10)$$

413 where $\langle \xi_z^2 \rangle$ is integrated strain variance, and h and j defined as

$$h(R_\omega) = \frac{3(R_\omega + 1)}{2\sqrt{2}R_\omega\sqrt{R_\omega - 1}} \quad (11)$$

$$j(f/N) = \frac{\text{arcosh}(N/f)}{f_{30}\text{arcosh}(N_0/f_{30})} . \quad (12)$$

414 where f_{30} is the inertial frequency at 30° latitude. Estimating the shear-to-strain ratio
 415 requires measurements of local temperature and salinity at higher resolution than avail-
 416 able from our moorings. We thus use a constant $R_\omega = 3$, the canonical Garrett-Munk
 417 value. Kunze et al. [2006] calculated a global value of $R_\omega = 7 \pm 3$ from CTD/LADCP
 418 observations, but the canonical value was found to be a reasonable estimate for the upper
 419 2000 m [Whalen et al., 2012]. Furthermore, the main goal of this analysis is to observe the
 420 vertical variability of the dissipation rate with respect to NI KE distribution. The strati-
 421 fication, N_{clim} , is derived from profiles of temperature and salinity from the WOCE-Argo
 422 climatology.

423 Enhanced turbulent dissipation rates are found in the permanent pycnocline for time-
 424 average ($\varepsilon \geq 1 \times 10^{-9} \text{ W kg}^{-1}$, Figure 5b) and a selected one-day average ($\varepsilon \geq 5 \times$
 425 $10^{-9} \text{ W kg}^{-1}$, Figure 5a) from the November 2019 NIW event. The vertical distribution
 426 of turbulent dissipation at the end of the NIW event mirrors the structure of NI KE
 427 with depth. (Figure 5a). Highest values for both turbulent dissipation rate and NI KE
 428 ($\sim 7.8 \times 10^{-9} \text{ W kg}^{-1}$ and $\sim 4 \times 10^{-3} \text{ m}^2 \text{ s}^{-2}$, respectively) are reached once NIW downward
 429 propagation has ceased and the NIW is trapped in the permanent pycnocline (Figure 5a).

430 Separating NI KE and ε averaged over the full time series by occurrences associated
 431 with negative vorticity show a similar pattern (Figure 5b, solid curves): enhanced turbu-
 432 lent dissipation rate with enhanced NI KE ($\sim 2.0 \times 10^{-9} \text{ W kg}^{-1}$ and $\sim 6.6 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$)
 433 in the permanent pycnocline. During occurrences of positive vorticity, NI KE and tur-
 434 bulent dissipation rate are not particularly high in the permanent pycnocline. Turbulent
 435 dissipation rates averaged for negative and positive vorticity are similar above and below
 436 the permanent pycnocline where they diverge from each other, a peak in dissipation be-
 437 ing observed only in anticyclones. Therefore, vertical propagation of NIWs promoted by
 438 anticyclonic vorticity and trapping in the permanent pycnocline are likely responsible for
 439 enhanced turbulent dissipation rates and associated diapycnal mixing in the permanent
 440 pycnocline.

441 Beyond the vertical structure of turbulent dissipation rates, we need to elucidate if

dissipation is the only reason why locally generated NI KE disappears from our observations after some time. This would allow to conclude on if NIWs are dissipated locally in the permanent pycnocline, thereby contributing to turbulent mixing in the area they were generated. If not, this would provide a first estimate of how other processes, such as energy transfers and horizontal propagation, could be at play to explain the disappearance of NI KE. Kunze et al. [2023] quantified the importance of local turbulent dissipation based on a dissipation time scale,

$$\varepsilon_t = \overline{NI KE} / \bar{\varepsilon} \quad (13)$$

where $\overline{X}$ represents the average of any variable X over the permanent pycnocline depth range (400 m - 900 m). The dissipation time scale is then compared to the period over which NI KE remains elevated in this layer during a NIW event, Δt (Figure 6). Three scenarios, labeled (A), (B) and (C), can be identified:

(A): If $\varepsilon_t = \Delta t$, then NI KE present in the permanent pycnocline is completely dissipated there.

(B): If $\varepsilon_t > \Delta t$, then turbulent dissipation is not sufficient to explain the weakening of NI KE in the permanent pycnocline. Other processes must be at play.

(C): $\varepsilon_t < \Delta t$ is not a scenario that should happen since it would mean that locally generated NI KE remains even though it should have been completely removed through dissipation from the permanent pycnocline.

Twelve NIW events were selected at mooring M1. They were identified using a threshold NI KE in the permanent pycnocline of $7 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$. ε_t and Δt were then computed for each event. Both variables have similar values with ε_t varying between 4 and 14 days and Δt between 4 and 20 days. Each estimate of Δt was estimated visually by isolating several events using a NI KE contour of $5 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$. Due to the numerous patterns of high NI KE in the permanent pycnocline, this method could not be generalized and Δt had to be estimated individually for each event, yielding an error of 2 days approximately. The error for ε_t was estimated using a bootstrap method for each event. Four events are found to follow scenario (A). The average dissipation time scale and Δt match (Figure 6, red square). This supports the notion that local dissipation of NI KE in the permanent pycnocline is the main factor for trapped NI KE disappearance at the mooring site. An

471 increase in NIW events coinciding with anticyclones would lead to an increase in turbulent
 472 mixing in the permanent pycnocline in Iceland Basin. For the remaining events, four follow
 473 scenario (B), with ε_t higher than Δt . Other mechanisms must be at play to explain the
 474 decrease in NI KE at the mooring location. The four remaining events fall under scenario
 475 (C), with ε_t lower than Δt . NI KE remains in the permanent pycnocline even after it
 476 should have been completely dissipated. This is due to the presence of a deep mixed
 477 layer. High values of NI KE in the permanent pycnocline are used to estimate Δt . Since
 478 the mixed layer overlaps with this layer for these two events, Δt includes NI KE in the
 479 mixed layer, i.e. inertial oscillation still forced, and therefore not completely dissipating.
 480 This effect can be enhanced by the presence of several events in a short period of time.
 481 This is the case for the longest event ($\Delta t \simeq 15$ days) in January. The event could also be
 482 characterized by several subsequent events, generating high trapped NI KE for a longer
 483 period.

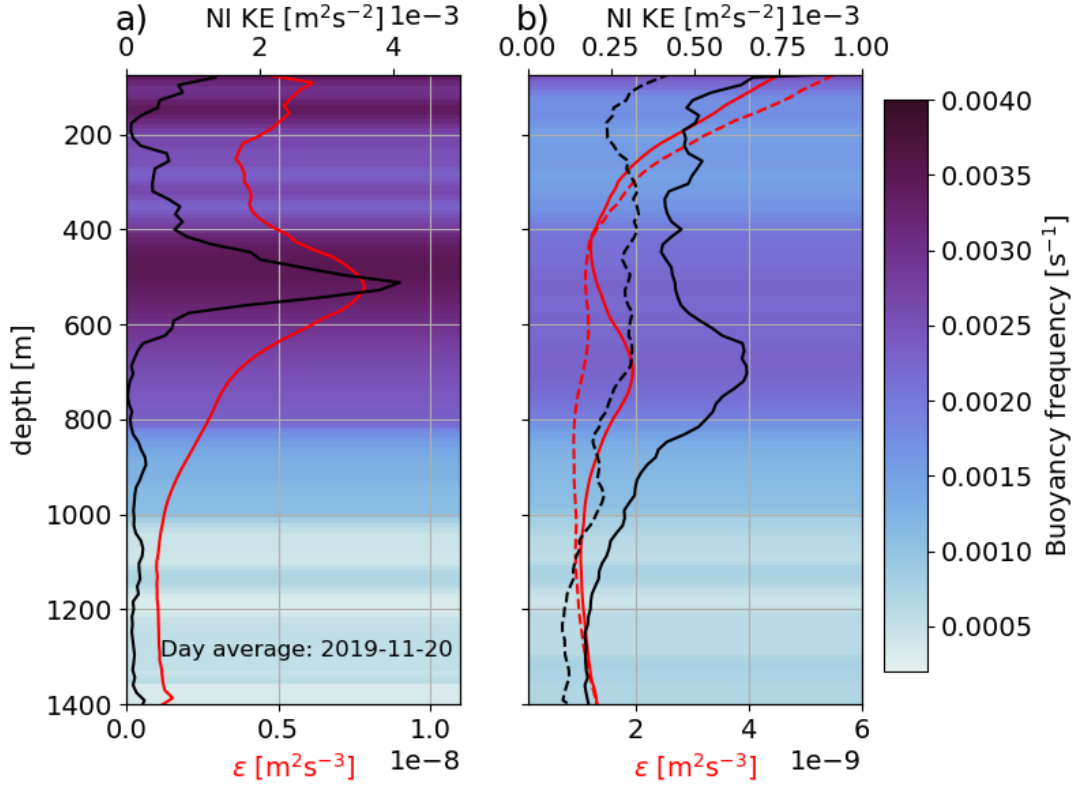


Figure 5: NI KE and turbulent dissipation rate ε vertical distribution. (a) One-day average of NI KE (black) and ε (red) at the end of the NIW event, on 2019-11-20 (Figure 3 right panels). (b) NI KE and ε averaged over the full time series depending on relative vorticity sign with solid curve corresponding to negative vorticity and dashed curve to positive vorticity. Background blue shades show corrected buoyancy frequency $N_{corrected}$ averaged over November 2019 (a) and buoyancy frequency N averaged over the whole year (b).

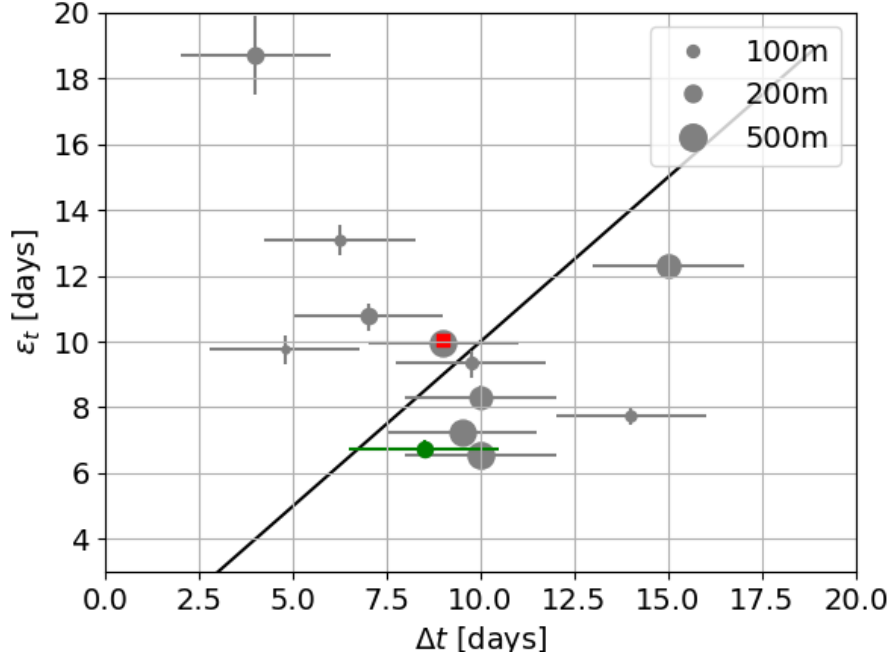


Figure 6: Dissipation time scale ε_t as a function of NIW event length Δt for eleven wind-driven NIW events at mooring M1. Time scales for each event are represented by gray circles. The overall mean value is plotted with the red square. The green circle represents the November event illustrated in Figure 3. The function $\Delta t = \varepsilon_t$ is represented by the solid black line. Error bars are plotted around each circle. Errors on Δt estimates were estimated around 2 days. Errors on ε_t were estimated using a bootstrap method. Circle area represents the mixed-layer depth averaged over Δt .

5 Summary and discussion

18 months of moored observations in Iceland Basin provide a comprehensive view of the life cycle of wind-driven NIWs from their generation in the mixed layer to their dissipation at depth. The three NISKINe moorings in a triangular array procured information on not only NIWs, but also the vertical mesoscale velocity structure, including vorticity. In particular, the moored time series highlight the NIW seasonal cycle and the role of background mesoscale motions and stratification for this cycle. Iceland Basin, and more specifically, the NISKINe mooring site, are particularly influenced by the generation of wind-driven NIWs and their downward propagation due to seasonal wind, mesoscale activity, and the presence of a permanent pycnocline. This pycnocline was located between 400 and 900 m during the observed period (June 2019 to October 2020) and is a permanent feature of the area [Kunze et al., 2023].

From September 2019 to April 2020, high wind stress coincides with predominant anticyclonic mesoscale currents, and a deepening of the mixed layer to 500 m. During this period, the moorings register the highest values of NI KE in the surface mixed layer but also in the permanent pycnocline below the mixed layer (on average 2.0 and 1.3 times the full time series mean NI KE at mooring M1, respectively). Increased NI KE at depth is linked to the presence of anticyclonic vorticity as high depth-averaged downgoing shear variance and NI KE below the mixed layer are observed in anticyclones. This process, known as the drainpipe effect, is caused by the shift to an effective inertial frequency in anticyclones [Thomas et al., 2020, Yu et al., 2022]. The vertical vorticity structure was estimated from the triangular mooring array. The seasonality of wind stress in Iceland Basin generating NIWs is known. However, the fact that high wind stress coincided with mostly anticyclonic mesoscale vorticity during winter 2019/2020 has not been reported before. The combination of wind and mesoscale is the necessary factor for downward propagation of NIWs, resulting in enhanced trapped NI KE in the permanent pycnocline. The formation of eddies in Iceland Basin has been reported to depend on the NAC path through the basin [Read and Pollard, 2001, Martin et al., 1998] and no previous studies showed a consistent intra-annual cycle of mesoscale vorticity. The presence of NI KE at depth should then also vary on longer time scales, when the suitable conditions are combined, which seems to have been the case for a few years during and before the NISKINe field work (Figure B4).

The downward propagation of NIWs stops in the permanent pycnocline and NI KE levels remain elevated at this depth for a few hours to a few days. This depth range also corresponds to a decrease in low-frequency kinetic energy and absolute vorticity. The combination of high stratification and anticyclonic mesoscale vorticity is the controlling element of vertical NIW propagation, stopping the downward propagation and trapping the waves in a fixed depth range. Once trapped, the fate of NI KE can follow different paths from dissipation to horizontal propagation outside of the mooring array range. We document the turbulent dissipation rate, which is an indication of where mixing by NIWs may occur. Highest turbulent dissipation rates are found in the surface mixed layer and in the permanent pycnocline ($\geq 3 \times 10^{-9} \text{ W kg}^{-1}$ and $\geq 1.5 \times 10^{-9} \text{ W kg}^{-1}$, respectively) during periods associated with negative vorticity. These are comparable with turbulent dissipation rate estimates by Kunze et al. [2023] ($1.2\text{--}7.0 \times 10^{-9} \text{ W kg}^{-1}$) based on profiling float observations in Iceland Basin. The trapping and dissipation complete the life cycle of the locally generated wind-driven NIWs. Unlike Kunze et al. [2023], we find that seasonal variability of turbulent dissipation rates at depth follows the seasonality of NI KE at the same depth. This indicates a combined influence of seasonal winds and mesoscale vorticity. We are able to disentangle this relationship because of the time resolution of the moored observations and the long duration of the deployment, resulting in the ability to accurately separate NIWs from semi-diurnal internal tides. Using estimates of dissipation time scales for selected events at mooring M1, we find most NIWs dissipate locally, in the permanent pycnocline. For a few events, however, turbulent dissipation rates are not sufficient to explain the disappearance of NI KE at depth. Other mechanisms must then be at play, including energy transfers or horizontal advection. This study provides us with a clearer picture of NIW life cycles and their complex interactions with both mesoscale and stratification. It opens several questions that, if answered, would shed a new light on these interactions.

One of these questions is the impact of NIW dissipation and associated turbulent mixing on the permanent pycnocline. This could lead to a decrease in buoyancy frequency as turbulent mixing takes place. Our observations indicate the permanent pycnocline remains consistent in magnitude during the periods of enhanced downward internal wave propagation, trapping and consequent dissipation of NIWs. However, permanent pycnocline stratification is visibly lower in summer 2020 than it was during summer 2019

(Figure B2). We find that NIWs generally dissipated locally which increases their potential of weakening the permanent pycnocline. We assess the impact of a single NIW mixing event on the permanent pycnocline by applying a one-dimensional model of vertical diffusivity [Hautala et al., 1996, Voet et al., 2015] to the density profile of the November 19 event (Fig. 5a). Vertical diffusivity κ is derived from the finescale turbulent dissipation rate estimate as $\kappa = \gamma \varepsilon / N^2$ assuming a dissipation flux coefficient $\gamma = 0.2$ [Osborn, 1980, Moum, 1996]. $\kappa \approx 10^{-3} \text{ m}^2 \text{ s}^{-1}$ for the dissipation peak around 500-m depth in Fig. 5a. Applying this κ to the permanent pycnocline over the course of 5 days reduces the stratification peak around 500-m depth by only about $5 \times 10^{-5} \text{ s}^{-1}$ or 1.5 %. We note that the relatively large uncertainty in stratification and its smoothness from using climatological data in the estimate influence both κ and the local pycnocline strength. Finding out more about the potential impacts of NIW-driven turbulent mixing on the long-term stratification would bring valuable information on how an increase in NIW activity, combined with the presence of anticyclones, could cause change in previously highly stratified areas such as Iceland Basin.

Investigating this issue further requires the use of higher spatial resolution datasets of stratification around the mooring array. With only five SBE 37s providing combined measurements of temperature and conductivity along mooring M1, our study was indeed limited due to our inability to obtain high-resolution local stratification throughout the water column at all moorings. Additionally, direct and long-term observations of turbulent dissipation will allow for accurate quantification of the impact of NIWs-driven turbulent mixing on the permanent pycnocline. An improvement of the stratification estimates, and/or the analysis of realistic numerical simulations would also serve the study of the role of stratification on the NIWs pathways. Reflection of NIWs on stratification jumps inside eddies has indeed been previously reported [Thomas et al., 2024]. Events of trapped NIWs could be isolated by combining moored observations, numerical simulations, and any profiling instruments available in the area to investigate this phenomenon.

Refining our knowledge of the complex interactions between eddies and NIWs is a domain to be improved in future studies. Part of future work includes disentangling the fate of trapped NIWs. This study focused on local dissipation of NIWs to investigate the cause for decreasing NI KE in the permanent pycnocline. An alternative mechanism is an energy exchange between NIWs and other motions, including mesoscale [Rocha

et al., 2018, Polzin, 2010, Kar and Barkan, 2023, Barkan et al., 2017, Dong et al., 2023]. NIWs could either gain or lose energy in the process. These transfers of energy can be studied by computing cross-scale/frequency energy transfers from moored velocity time series [Garabato et al., 2022]. In our case, a transfer of energy from trapped NIWs to anticyclones could explain part of the NI KE decrease once trapped in the permanent pycnocline (Figure 3). High-resolution simulations could be explored to test the accuracy of energy transfer estimates from a coarse mooring array. Energy transfers between motions at different frequencies can be estimated using a coarse-graining method. However, this requires the estimation of horizontal gradients of the velocity fields [Aluie et al., 2018, Garabato et al., 2022]. Such estimates can be obtained from a triangular array using interpolation methods or averaging, but the methodology efficiency should be carefully assessed using numerical simulations.

Another explanation for the decrease and/or disappearance of trapped NI KE is horizontal propagation of NIWs outside of the mooring site. For NIWs trapped in an anticyclone, this horizontal propagation would follow the route of the eddy. Tracking anticyclones and trapped NIWs in Iceland Basin is not possible from moored observations. However, using a combination of satellite altimetry and surface drifters, this may be possible. Sea surface height provided by reanalysis products provide global coverage but their horizontal resolution does not resolve well small mesoscale or sub-mesoscale eddies. New satellites, such as SWOT (or future ODYSEA mission aiming to directly measure velocity fields), can provide higher resolution measurements, allowing for improved eddy tracking. These datasets, combined with surface drifters, may inform about the path followed by mesoscale eddies passing through the NISKINe region, providing improved EKE and vorticity estimates. Combined with available satellite data and numerical simulations, the NI KE distribution in anticyclones depending on distance to the center and bottom of an eddy could be studied along with the evolution of NIW modal composition. Such study would explore an aspect of NI KE horizontal distribution in Iceland Basin beyond the scope of the results presented in our study. Moored observations deployed as part of NISKINe provided valuable information on the vertical modulation of the NIW life cycle, including seasonal variability, within a region of active mesoscale and seasonally variable stratification. It enabled the description of the statistical significance of NIW propagation and trapping in anticyclones and consequent turbulent dissipation in a highly stratified

612 depth range.

613 **Acknowledgments**

614 This study was carried out through funding from the Office of Naval Research via grants
615 N00014-18-1-2423 and N00014-22-1-2182. We thank captains Derek Bergeron and Kent
616 Sheasley and the whole crew of R/V Neil Armstrong for outstanding support of our
617 science operations at sea in 2019 and 2020. We are grateful for UNOLS, ONR, and NSF,
618 facilitating a safe joint OSNAP/NISKINe mooring recovery cruise during the early days
619 of the COVID-19 pandemic. We thank John Colosi and Harper Simmons for adding
620 instruments to the moorings.

621 **Data availability statement**

622 NISKINe moorings data are available at <https://doi.org/10.5061/dryad.g1jwstr3w>.
623 Sea level derived from satellite altimetry are available on the E.U Copernicus Marine Ser-
624 vice website, <https://doi.org/10.48670/moi-00148>. Wind speed was obtained from
625 ERA5 reanalysis, <https://doi.org/10.24381/cds.adbb2d47>. Finally stratification was
626 derived from the World Ocean Circulation Experiment (WOCE)-Argo climatology tem-
627 perature and salinity, https://doi.org/10.1594/WDCC/WAGHC_V1.0.

Appendix A

Methods

Extracting near-inertial waves

This study is based on analysis resulting from observations obtained in the eddy-rich region just south of Iceland, away from any large topographic features. The nature of the observations, i.e. sparse spatial coverage, as well as the choices made during processing introduce potential biases which have been carefully assessed. We discuss several of the caveats here.

NIWs were extracted from moored horizontal velocity time series using a band-pass filter with fixed bandwidth and a central frequency. These two parameters control the fraction of the NI signal that will be extracted. The choice of the bandwidth was based on mean frequency spectra at mooring M1 (Figure A1). It was meant to extract near-inertial motions while excluding as much of the semi-diurnal tidal signal as possible. Seasonal variations of semi-diurnal internal tides and NIWs were therefore not taken into account by the filter. In summer (June to August), M_2 tides are predominant compared to the inertial peak (Figure A1, green curve), and the possibility of contamination of the near-inertial signal increases. On the contrary, during winter (December to February), the inertial peak dominates (Figure A1, orange curve). The latter season is indeed the one during which most of the NIWs were observed. The choice of the bandwidth and the seasonality of NIWs kept the possible contamination by M_2 tides to a minimum and no impact on the results presented in this study should be observed. This is further supported by the study of the filtered and demodulated velocity field (not shown). Assuming that the filtered signal contains two components with different frequencies, M_2 and f , the signal can be simply described as the superposition of the two oscillating signals, with constant frequency, $u_{M2}e^{i\omega_{M2}t} + u_f e^{ift}$. The demodulation of this signal by multiplying by e^{-ift} would introduce oscillations of frequency $\omega_{M2} - f \simeq 0.24$ cpd. This artifact of the filtering method was not observed in our case.

The central frequency of the filter is the other parameter controlling the effectiveness of NIW extraction. In this study, it was chosen to remain fixed at local Coriolis frequency, f . This was mainly devised to keep the central frequency away from the semi-diurnal peak. However, this choice could cause an underestimation of NI KE. In the presence

of ambient vorticity ζ , the near-inertial wave frequency is shifted from f to an effective inertial frequency, f_{eff} ,

$$f_{eff} = f + \zeta/2 . \quad (14)$$

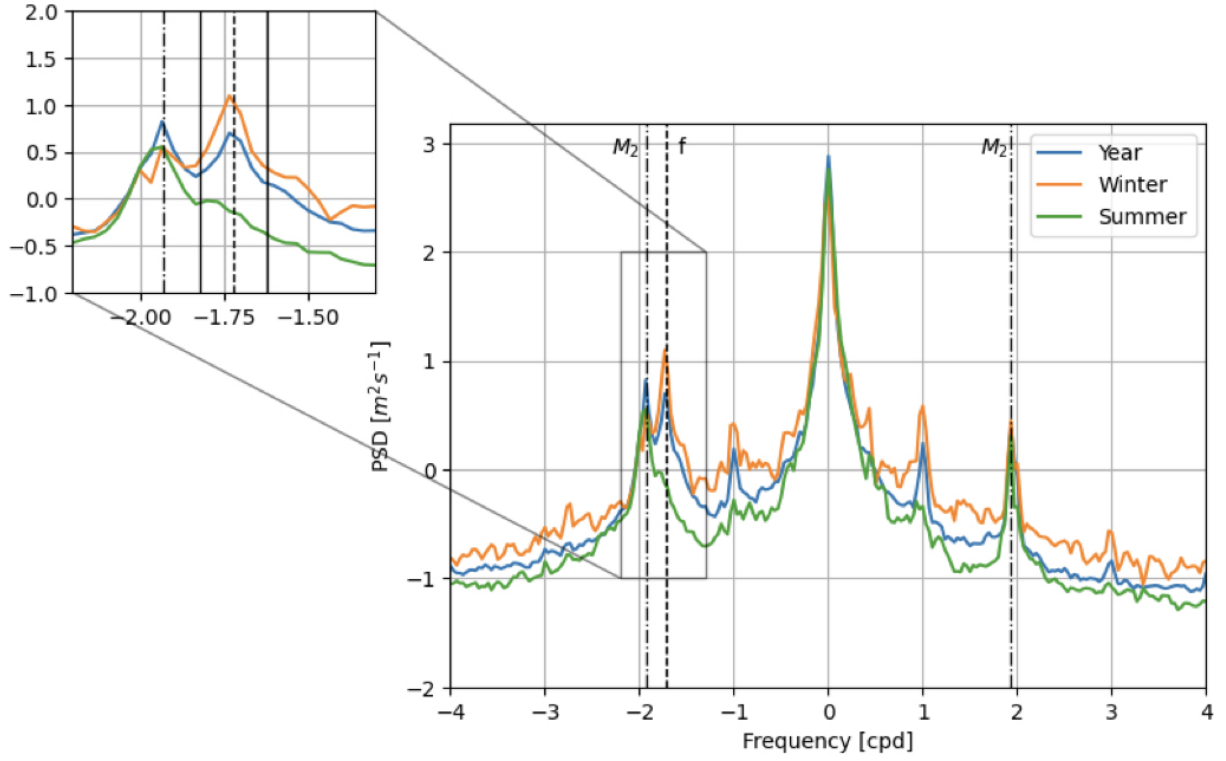
The frequency of an NIW energy peak will then depend on ambient vorticity sign and amplitude as presented in this study. The frequency shift was observed in the signal (Figure A1, orange and green curves), and choosing f as central frequency can cause an underestimation of NIW velocities and associated NI KE. This underestimation was estimated by using a theoretical study of the frequency power spectra (PSD) [Caspar-Cohen, 2022]. The theoretical PSD of an oscillating exponentially decorrelated signal [Sawford, 1991] can be described as

$$PSD(\omega) = \frac{T U^2 (T^2 (\omega + f_{eff}^2) + 1)}{T^4 (-\omega^2 + f_{eff}^2)^2 + 2T^2 (\omega^2 + f_{eff}^2) + 1} \quad (15)$$

where T is the decorrelation time scale and U the signal variance. The filtering of a signal by band-pass filtering, following the method presented in section 2, can be roughly compared to the integral of the PSD between two given frequencies, centered on f , as a perfect filter should produce the same amount of energy. The underestimation of NI KE caused by the use of constant f instead of time-varying f_{eff} can then be estimated by comparing the integration of (15) in two frequency bands: $f \pm 0.1$ cpd (used in this study) and $f_{eff} \pm 0.1$ cpd. Integration over the band centered on f_{eff} provides what we will refer to as the true NI KE. An underestimation of the NIW signal is indeed observed, however, it depends on the absolute vorticity, and on average remains small (underestimation represents 9% of the averaged true NI KE). This theoretical study was carried out without taking into account the semi-diurnal tides. Depending on the season, the use of f_{eff} could also lead to an overestimation of the NI signal. A complete study would be necessary to choose a "perfect" filter, as the bandwidth and central frequency should depend on ambient vorticity estimates, but also decorrelation time scales for both tidal and near-inertial signals. Based on the observations and the further theoretical analysis, the simpler filter with fixed bandwidth and central frequency was found to have no significant impact on the object of our study: the NI KE vertical distribution. Temporal and spatial evolution of enhanced NI KE were successfully highlighted, even during periods of high ambient vorticity.

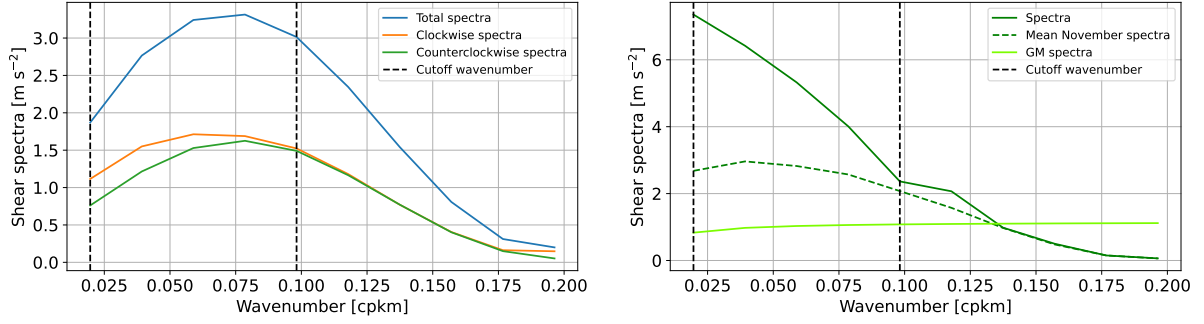
Stratification estimation

Apart from signal processing choices, another caveat appeared during our study: the lack of conductivity measurements along the entire mooring depth. CTD data were recorded at only 5 depths at mooring M1. Local stratification N was therefore estimated based on temperature, measured at higher resolution along the mooring, and WOCE-Argo climatological salinity profiles, interpolated to the mooring location. Comparison of local N and climatological N_{clim} time-averaged stratification profiles shows similarities with the permanent pycnocline observed in both datasets. The average permanent pycnocline is, however, found at slightly different depth ranges, with the locally estimated pycnocline shallower than the climatological one (400-850 m locally and 500-950 m in the climatology). Stratification profiles averaged over shorter periods show larger differences than the annual average. While the permanent pycnocline remains visible, large intrusions were found in monthly averages of N . These intrusions are most likely caused by the absence of local salinity observations to balance temperature profiles. In the $N_{corrected}$ estimates averaged over November 2019 (section 4.3), we used data from 3 CTDs as reference to get "true" local stratification at three depths (157 m, 338 m and 606 m on average). These reference estimates are then used to remove $N_{corrected}$. Looking at N averaged over November 2019, we observed a large intrusion where the permanent pycnocline was expected to be. Using $N_{corrected}$, we found this was an artifact introduced by the stratification estimation method. The monthly average was then smoothed accordingly, using the CTD-based estimates as the "true" estimates. As long as the stratification profiles are averaged over long time period, as done in our study, the estimation method using only local temperature works well and allows to define the permanent pycnocline depth range. However, it could hinder the study of shorter events by producing inaccuracies in the localization of the permanent pycnocline.

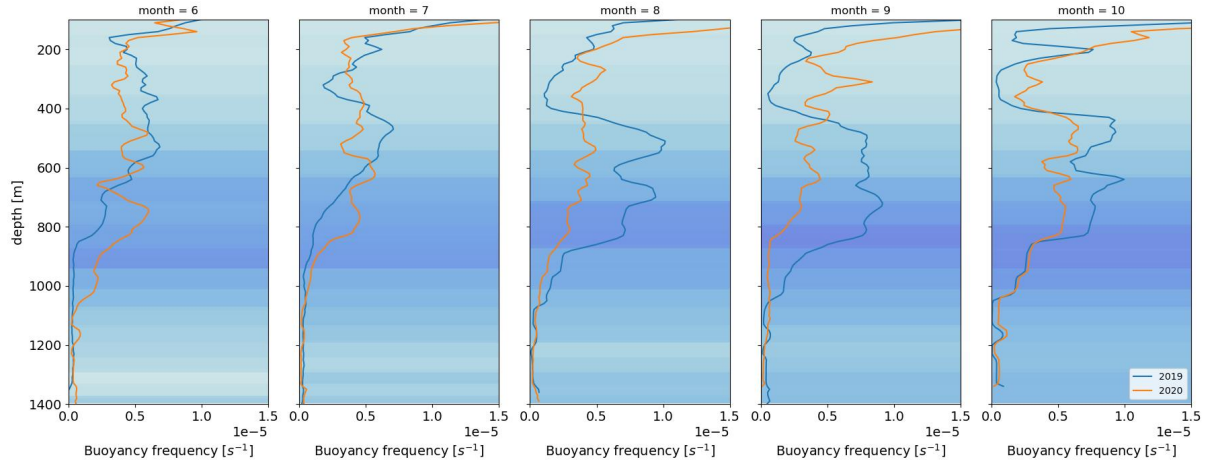


A 1: Depth-averaged velocity rotary spectra and filter frequency band. (Right) Velocity rotary spectra computed using the full 18-month time series from mooring M1, winter months (JFM), and summer months (JJA). Coriolis and semi-diurnal frequencies are indicated by dashed lines. (Left) Zoom on near-inertial and semi-diurnal frequency peaks. Limits of the frequency band used in the band pass filter are described in section 3.4.

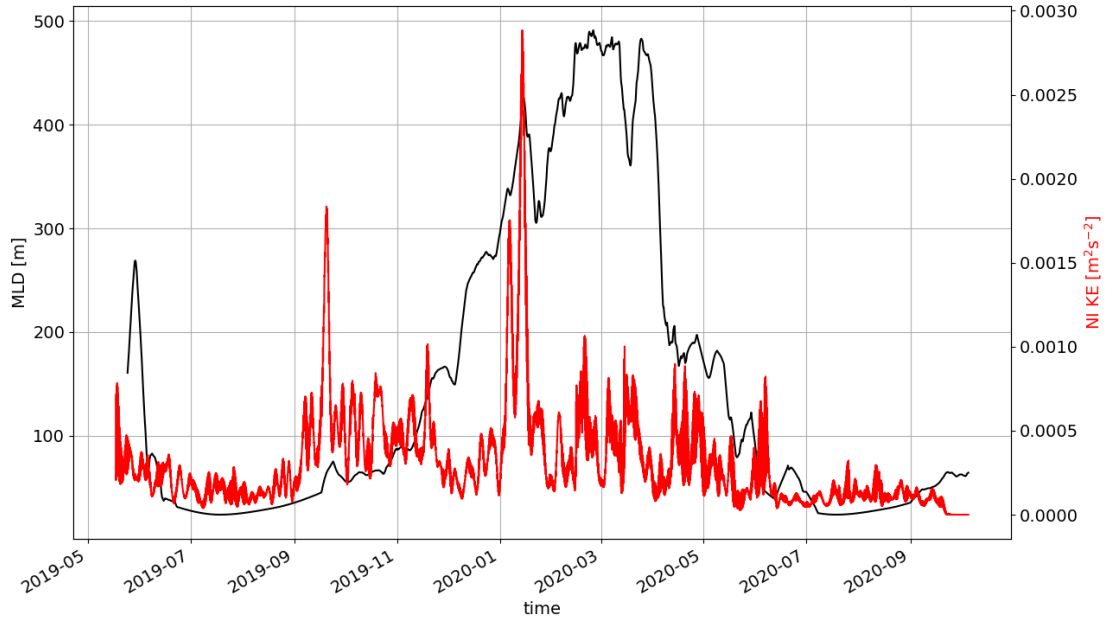
Appendix B: Figures



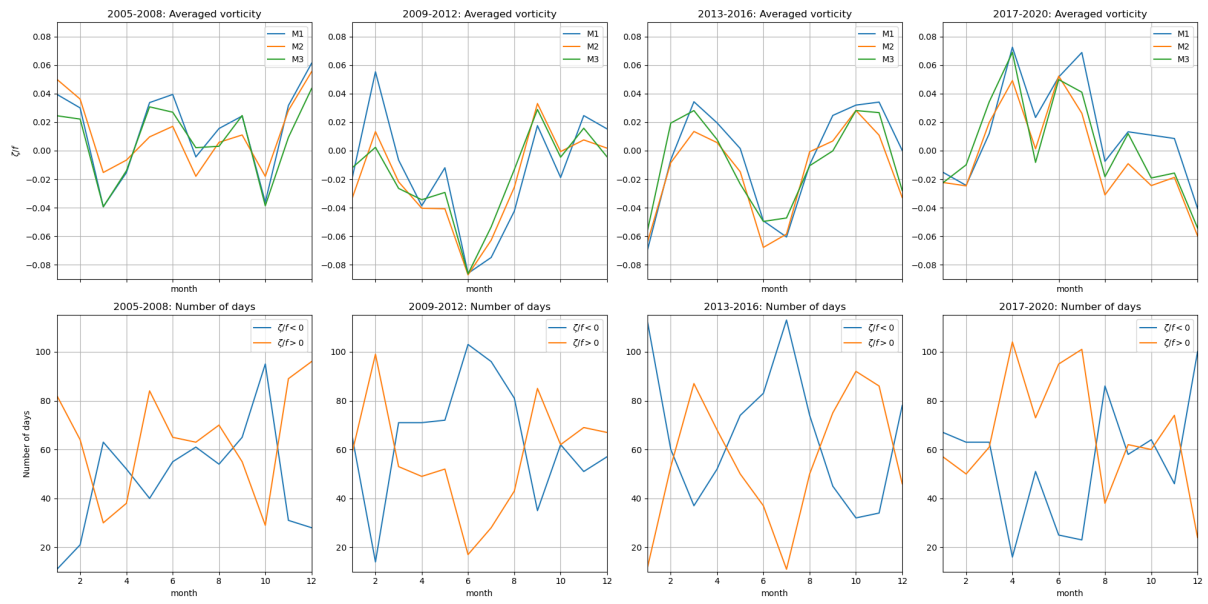
B 1: Depth- and time-averaged vertical shear spectra and cutoff wavenumbers (dashed black lines) used to derive dissipation rates from velocity fields. (Left) Averaged spectra over the entire year and depth range. Clockwise (orange curve) and counterclockwise (green curve) components of the averaged shear spectra, as well as the total averaged spectra (blue curve), defined as sum of the two, are represented. (Right) Spectra averaged over a day (solid dark green curve) and November 2019 (dashed dark green curve) as well as the Garrett-Munk theoretical spectrum (light green curve) are plotted.



B 2: Monthly average buoyancy frequency N from June to October. Averages for 2019 (blue curves) and 2020 (orange curves) are compared. Background represents the monthly averaged climatological stratification N_{clim} profiles. The background colormap ranges from $1 \times 10^{-6} \text{ s}^{-1}$ to $5 \times 10^{-5} \text{ s}^{-1}$.



B 3: Smoothed mixed-layer depth (black curve) and NI KE averaged below MLD (red curve).



B 4: Monthly evolution of satellite altimetry-derived surface vorticity (top panels) and sign (bottom panels) over 4 time periods of 4 years each from 2005 to 2020. (Top) Vorticity is averaged monthly for each mooring. (Bottom) Number of days during which vorticity is either positive (orange) or negative (blue) at M1 for each month.

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